

Runaway Probability Neural Model Development

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Abstract

This report documents development of a JAX-trained neural model for the steady relativistic runaway-electron first-passage probability. The project begins with a vanilla supervised model and adds architectural, numerical, and physics-informed features only after each simpler stage is measured.

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1 Purpose and terminology

The target is a bounded response function

$$0 \leq P(p, \xi; \boldsymbol{\lambda}) \leq 1, \quad (1)$$

where p is dimensionless momentum, ξ is pitch cosine, and $\boldsymbol{\lambda}$ contains plasma and field parameters.

Neural model is the general term used in this report. JAX provides the numerical and automatic-differentiation framework used to train models. *Physics-informed neural network* (PINN) refers only to a training mode that includes a physics residual or boundary constraints in its objective. Supervised data-driven training is a neural-model baseline, not automatically a PINN.

2 Scientific problem

The model estimates first-passage probability for an electron evolving in a relativistic kinetic system. The successful event is exit through the upper momentum boundary; failure is exit through the lower-momentum boundary. The response depends on seven physical parameters:

$$\boldsymbol{\lambda} = (E/E_c, T_e, n_D, n_{Ne}, z_D, z_{Ne}, B). \quad (2)$$

The CPU finite-volume solver defines reference labels. Neural models learn an approximation to those labels or to the governing normalized residual, depending on training mode.

3 Reference finite-volume data

The reference solver uses a conservative finite-volume discretization of the kinetic operator and a volume-weighted discrete adjoint. The dataset pipeline performs the following operations:

1. Generate reproducible Sobol parameter cases.
2. Build composite momentum grids with upper-tail concentration for near-threshold cases.
3. Solve the sparse adjoint first-passage system on CPU.
4. Add an analytic zero-probability low-momentum tail.
5. Reject trivial or invalid cases.
6. Store exact cell centers, fields, metadata, and checksums.

Finite-volume labels remain outside JAX automatic differentiation. This separation keeps reference-label generation independent from model training. The current dataset contains 131072 cases on a coarsened 232×44 phase-space grid. Boundary-adjacent cells are retained during coarsening so the upper-momentum success boundary is not discarded.

4 Normalized model interface

The pointwise model input has nine normalized coordinates:

$$\mathbf{z} = (z_p, z_\xi, z_{E/E_c}, z_{T_e}, z_{n_D}, z_{n_{Ne}}, z_{z_D}, z_{z_{Ne}}, z_B). \quad (3)$$

The first two coordinates represent phase space. The remaining seven represent the physical parameter case. Logarithmic mappings are used for positive quantities where appropriate. All model calculations use FP64.

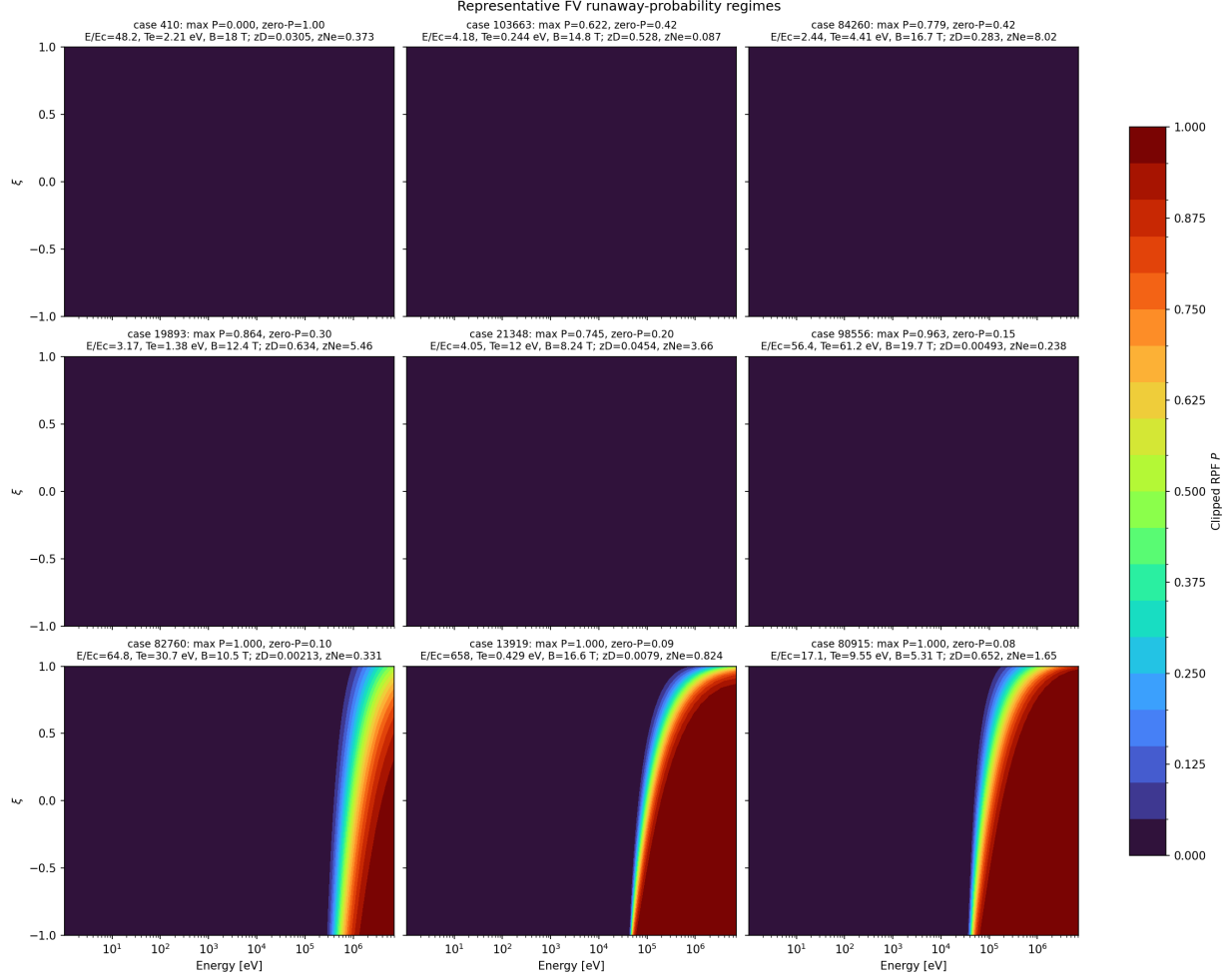


Figure 1: Representative clipped FV RPF solutions across selected physical regimes.

4.1 Initial production domain

The initial production configuration defines the following parameter ranges. The JSON files remain operational run definitions; this table records the domain used by the current model-development baseline.

Parameter	Minimum	Maximum	Mapping
E/E_c	1	1000	logarithmic
T_e [eV]	0.1	100	logarithmic
n_D [m ⁻³]	10 ²⁰	10 ²²	logarithmic
n_{Ne} [m ⁻³]	10 ¹⁶	10 ²²	logarithmic
z_D	0.001	1	logarithmic
z_{Ne}	0.001	10	logarithmic
B [T]	1	20	linear

The phase-space domain uses $p \in [0.002, 15]$ and $\xi \in [-1, 1]$. Charge-state means use $z_D, z_{Ne} \geq 0.001$ to sample sub-unity states without reaching zero-electron singularities. Domain validation

enforces positive ordered bounds and matching parameter ranges across FV and neural-model configurations.

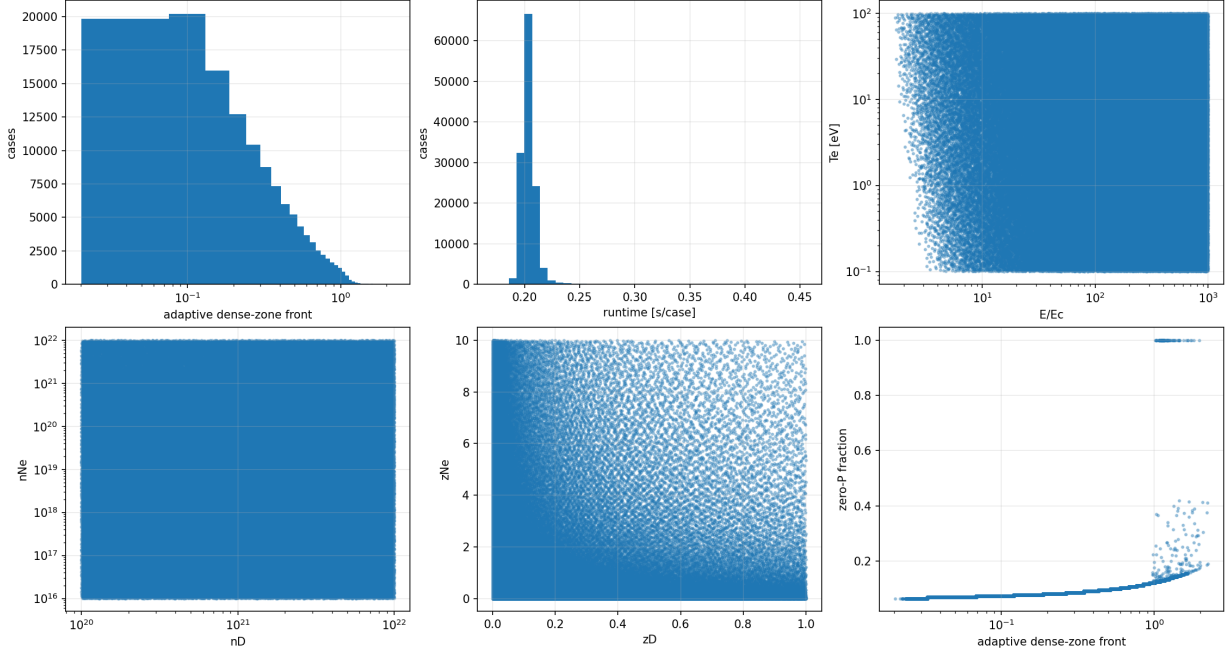


Figure 2: FV dataset coverage and solver-runtime diagnostics generated by the dataset analysis routine.

5 Model development sequence

Development proceeds from the smallest useful baseline toward more structured models:

1. **Vanilla MLP.** Train a pointwise MLP with supervised FV labels.
2. **Baseline measurements.** Record train/test error, regional error, probability bounds, and reproducibility across seeds.
3. **Output constraints.** Compare bounded output transforms and low-momentum behavior.
4. **Architecture growth.** Test wider/deeper MLPs and, when useful, a DeepONet representation separating parameter and phase-space inputs.
5. **Physics-informed training.** Add PDE, threshold, and boundary terms only after the supervised baseline is understood.
6. **Active acquisition.** Use residual ranking to request new CPU FV labels only after fixed-data behavior is qualified.

Each stage must preserve a clear comparison with the previous stage. Added complexity is justified by measured improvement, not by architecture alone.

6 Initial vanilla experiment

The first experiment uses supervised FV data with a pointwise MLP. The model has width 128, six hidden layers, nine input coordinates, and a sigmoid output transform. SOAP uses $\beta_1 = \beta_2 = 0.997$, zero weight decay, an initial learning rate of 10^{-3} , cosine decay to a final fraction of 0.1, and a global batch size of 131072.

Training uses an 80/20 parameter-case split, 5000 optimizer steps, and 16 A100 GPUs across four nodes. Checkpoints are written at steps 1000, 2000, 3000, 4000, and 5000; the loss history also records the initial step.

The run uses 104858 training cases and 26214 test cases. Initial results are:

Metric	Value
Training MSE	7.84742×10^{-5}
Test MSE	1.53796×10^{-4}

These results define the first baseline for later architecture and physics-informed comparisons.

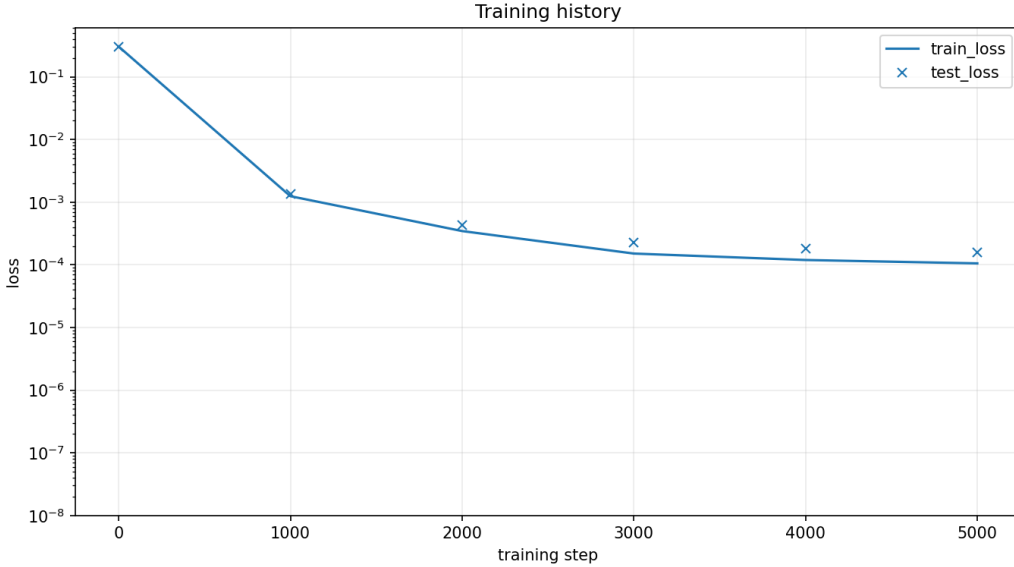


Figure 3: Training and test loss history for the vanilla MLP.

6.1 Validation results

The trained model was evaluated against all 131072 generated FV cases, or 1337982976 exact FV cells. Validation used the completed model and dataset manifests before computing predictions and residuals.

Metric	Value
Validation MSE	1.00219×10^{-4}
Maximum absolute error	9.87202×10^{-1}
PDE residual RMS	4.28710×10^{-2}

The regional MSE is 1.45221×10^{-5} in the zero-probability region, 1.79634×10^{-4} in the transition region, and 3.56743×10^{-7} in the saturated region. The largest errors occur near the transition boundary.

7 Data-plus-physics experiment

The second experiment uses the corrected composite-grid dataset while keeping the same MLP, SOAP optimizer, and 16-GPU allocation, then adds the PDE residual to the data objective. The PDE loss uses the same 65536 training FV input coordinates, so no separate collocation set is introduced. The data-loss weight is 1 and the PDE-loss weight is 10^{-3} ; threshold, low-momentum, and high-momentum boundary losses remain disabled.

Metric	Value
Training MSE	6.11134×10^{-4}
Test MSE	1.83293×10^{-3}
Validation MSE	1.73581×10^{-3}
Validation maximum absolute error	9.98960×10^{-1}
Validation PDE residual RMS	5.47506×10^2

This weighting substantially worsens supervised prediction relative to the data-only baseline. The result indicates that PDE weighting and residual normalization need tuning before drawing a physics-benefit conclusion.

8 Earlier data-plus-physics with exact upper-boundary loss

The earlier third experiment retained the data-plus-PDE setup and enabled the exact upper-momentum success boundary loss. It uses 16384 boundary collocation points, boundary-loss weight 1, data-loss weight 1, and PDE-loss weight 10^{-3} . This run predates the corrected composite-grid dataset and has not been rerun here. The boundary targets are evaluated at $p = p_{\max}$ using the outward-drift success condition.

Metric	Value
Training MSE	2.75380×10^{-3}
Test MSE	3.59397×10^{-3}
Validation MSE	3.46047×10^{-3}
Validation maximum absolute error	9.98953×10^{-1}
Validation PDE residual RMS	1.12364×10^2

This boundary-weighted configuration does not improve the initial hybrid baseline. Boundary-loss weighting and PDE normalization require further tuning before this objective can be judged beneficial.

9 Training modes

9.1 Supervised data mode

The model minimizes error between neural predictions and FV probabilities. Cases split at parameter-case level, not individual cells, so test cases measure interpolation across physical parameter space.

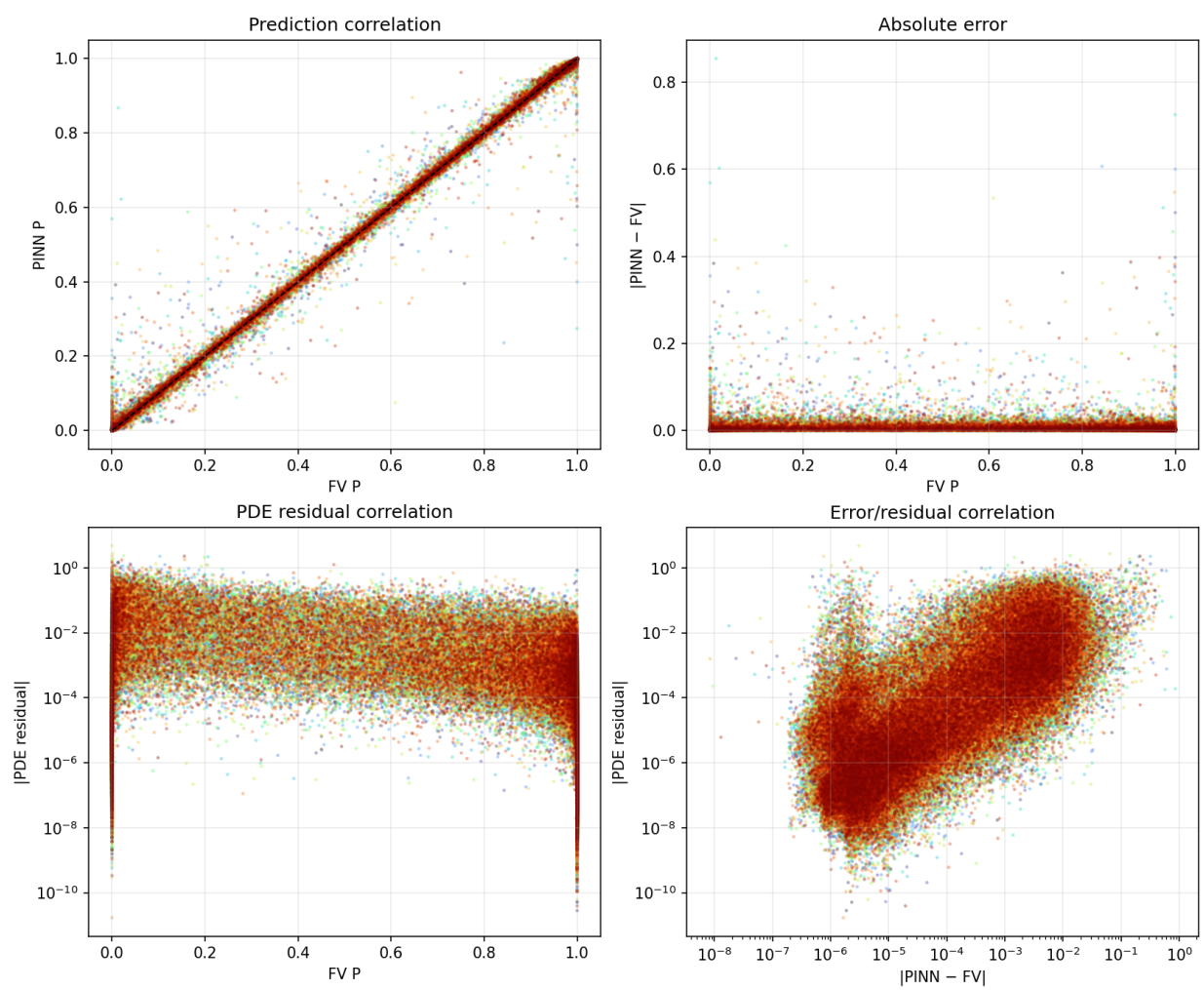


Figure 4: Vanilla MLP validation correlation, absolute error, and PDE residuals.

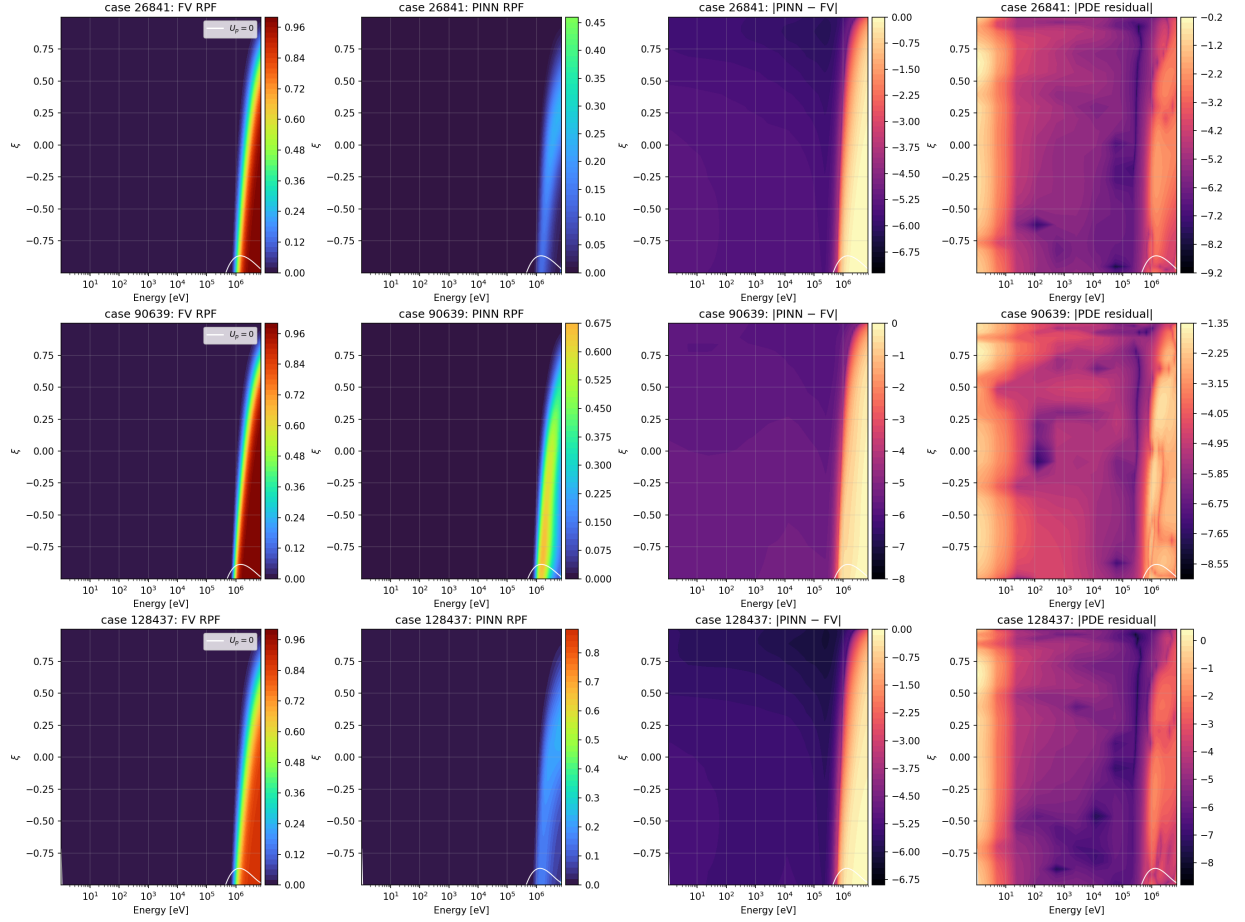


Figure 5: Representative worst-error validation cases from the vanilla MLP.



Figure 6: Training and test loss history for data-plus-physics training.

9.2 Physics-informed mode

The model objective may combine FV data loss, normalized PDE residual loss, threshold residual loss, low-momentum loss, and upper-boundary success loss. The term “PINN” applies to this mode because automatic differentiation contributes physics residuals to training.

9.3 Active training

The model ranks candidate points by PDE residual. Selected parameter cases are solved by the CPU FV pipeline, then appended to the training data. FV acquisition remains outside JAX differentiation.

10 Evaluation protocol

Every model stage should report:

- global MSE and maximum absolute error;
- error in zero, transition, and saturated probability regions;
- PDE residual statistics when physics terms are active;
- per-case metrics on held-out parameter cases;
- model, dataset, configuration, and environment identifiers.

Validation must verify completed manifests and checksums before analytics. Production generation, training, and validation run inside Slurm allocations.

11 Reproducibility contract

Production runs use the project-local environment, explicit JSON configuration, fixed Sobol seeds, FP64 JAX settings, fresh artifact paths, and atomic manifest publication. Generated datasets, models, checkpoints, histories, and validation outputs are run artifacts rather than source files.

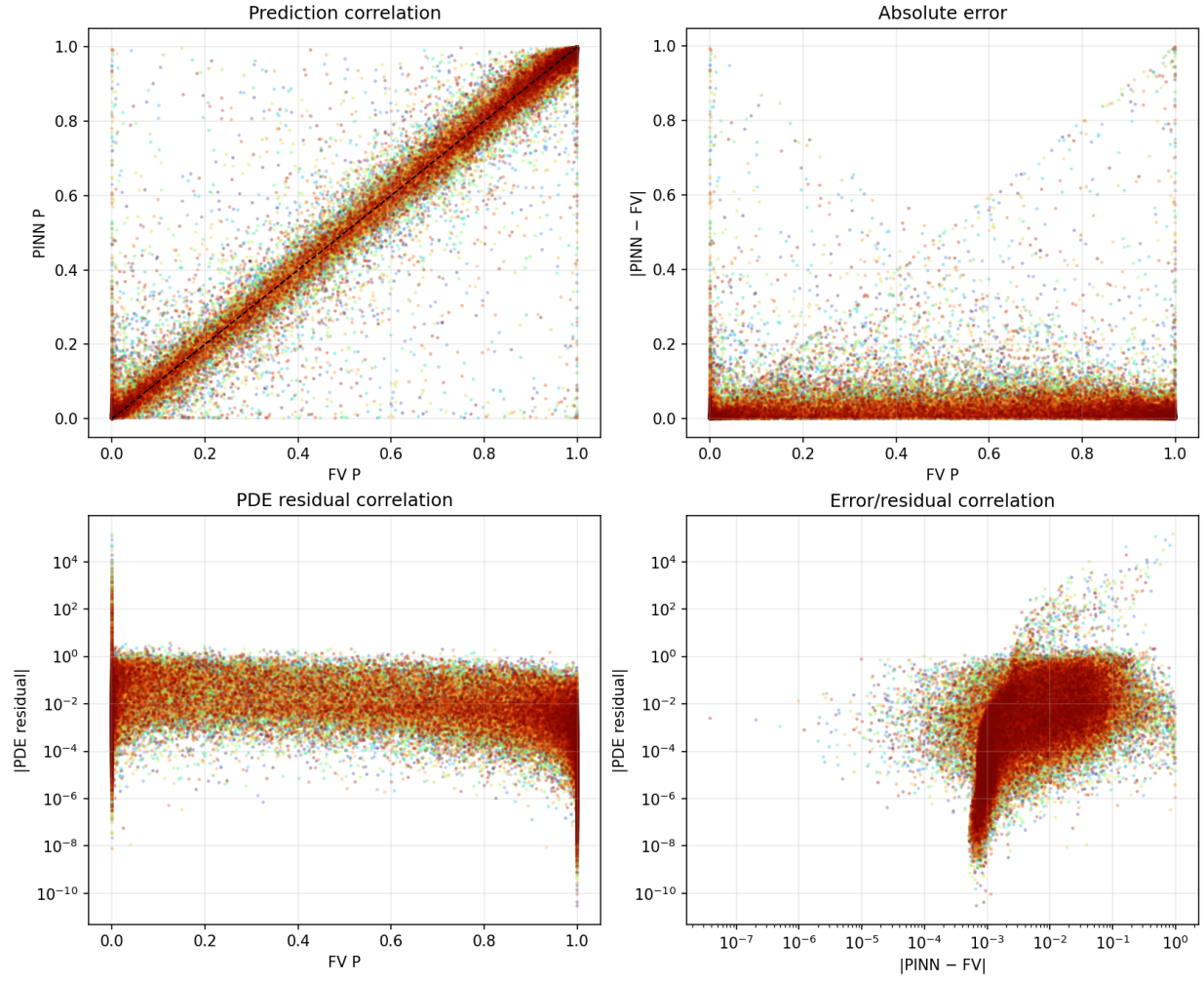


Figure 7: Data-plus-physics validation correlation, absolute error, and PDE residuals.

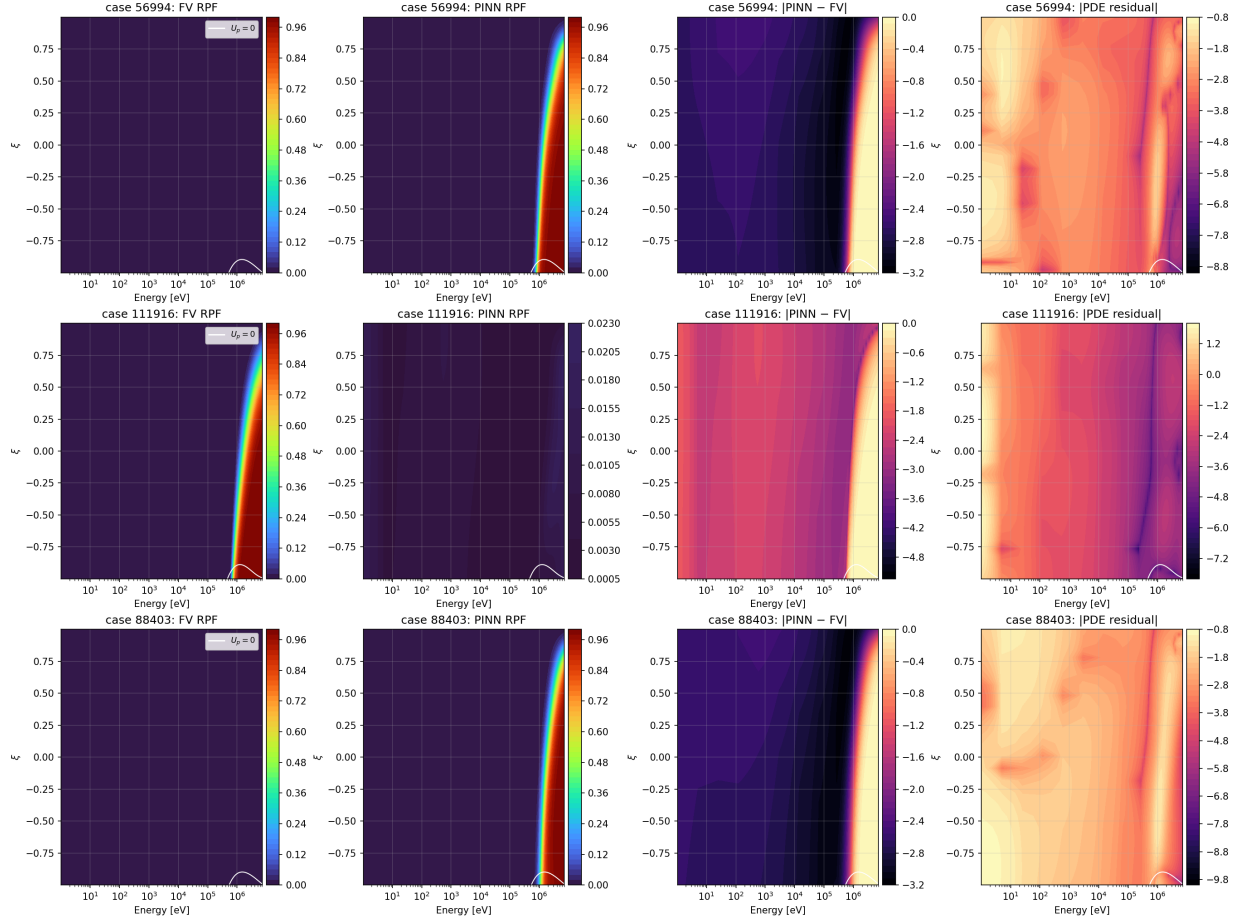


Figure 8: Representative worst-error validation cases from data-plus-physics training.



Figure 9: Training and test loss history with upper-boundary loss.

12 Current scope and next steps

The immediate objective is a trustworthy vanilla supervised MLP baseline. Later sections will record experiments, model changes, loss weights, compute resources, and validation results. Physics-informed training, DeepONet architectures, active acquisition, and SSBroyden refinement remain extensions to evaluate against that baseline.

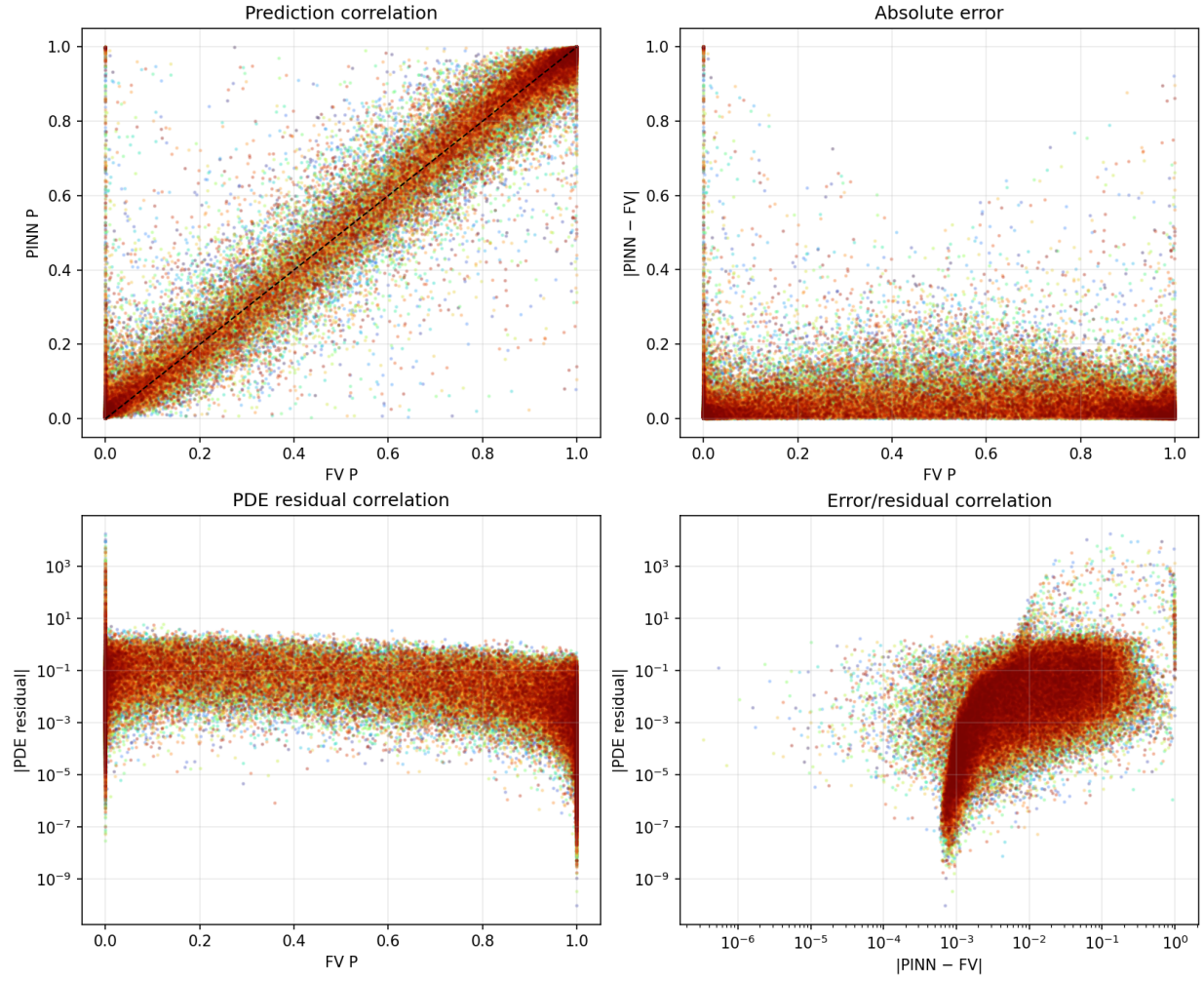


Figure 10: Boundary-loss validation correlation, absolute error, and PDE residuals.

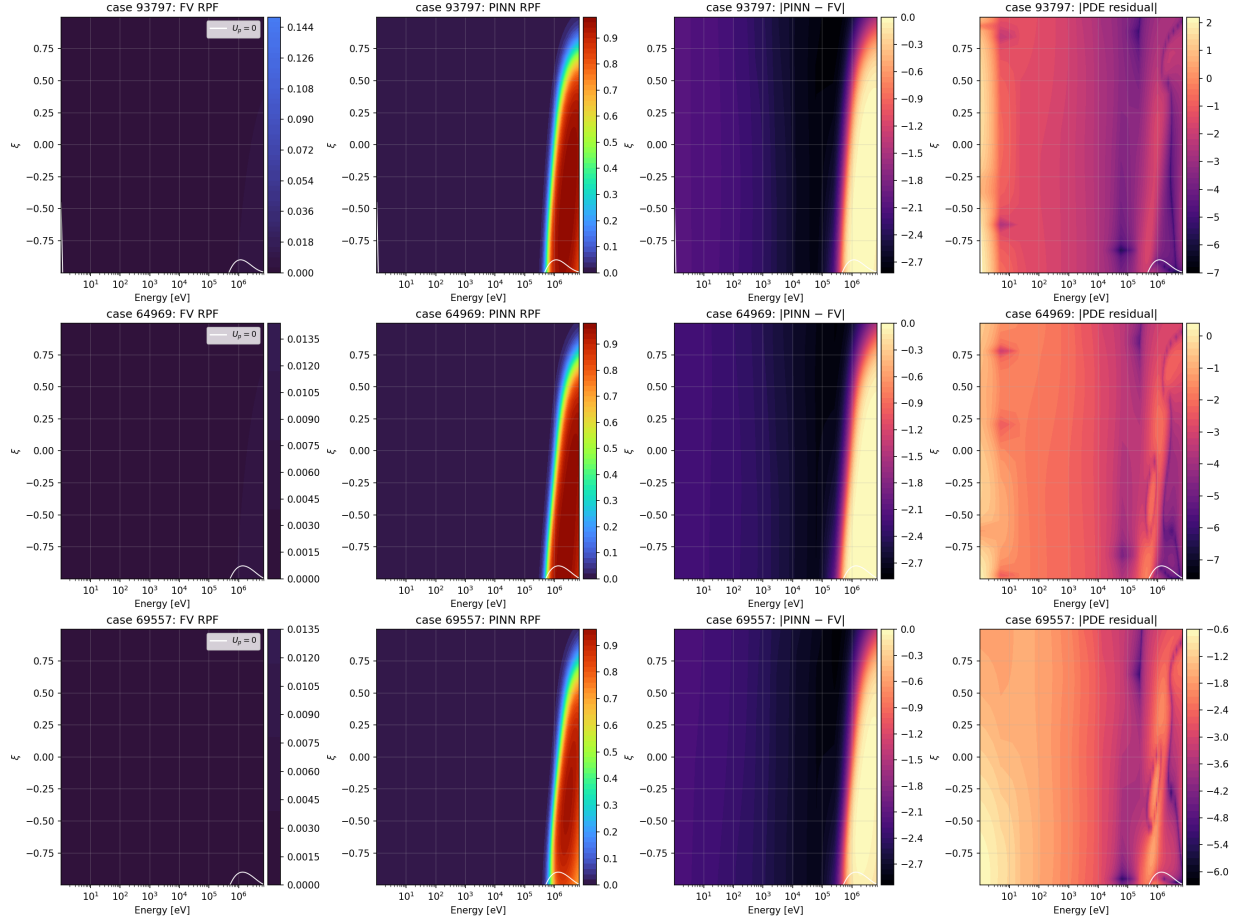


Figure 11: Representative validation cases with upper-boundary loss.